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Electricity Consumption in West Bengal: An Exploratory Analysis

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Abstract

This study explores how electricity consumption in West Bengal changed between 2002 and 2022. Starting with a basic analysis of annual data, it identifies overall growth trends and fluctuations over the years. A simple linear trend model is used to capture the long-term pattern, explaining most of the variation in usage. To reflect the drop in consumption during the COVID-19 years (2020–21 and 2021–22), a dummy variable is added. This decline is likely tied to reduced commercial and industrial activity during lockdown. The model’s reliability is supported by residual and seasonal analyses, and it achieves a strong R-squared value of 0.96. While alternative models like quadratic and piecewise trends were briefly tested, the linear model with the COVID adjustment turned out to be both clear and effective. Overall, the model does a solid job of capturing long-term trends while accounting for the temporary effects of the pandemic.

1 Introduction

Electricity use tells us a lot about how a region is growing and changing: economically, socially, and even culturally. In West Bengal, where both industrial development and urban growth have picked up in recent years, understanding how electricity consumption has shifted over time can offer useful insights. It helps answer questions about how demand has grown, whether usage patterns have been stable, and how external events may have caused sudden changes.

This report looks at electricity consumption in West Bengal from 2002 to 2022, not just to describe the changes, but to model and analyze them statistically. By exploring long-term trends and identifying unusual shifts, the goal is to see how well simple trend models can explain the data. In doing so, the study also considers how certain disruptions—like the COVID-19 pandemic—affected overall demand. This kind of analysis is especially helpful for planning ahead, improving forecasts, and making smarter energy decisions.

2 Data Exploration

The dataset used in this analysis was sourced from the Central Electricity Authority's official website. It contains figures for per capita electricity consumption, measured in kilowatt hours (kWh), for all Indian states throughout fiscal years 2002-2003 to 2021-22. For the purposes of this study, only the data corresponding to West Bengal has been considered.

The dataset provides annual values, making it suitable for long-term trend analysis. Since the values are reported on a per capita basis, they account for changes in population and offer a more standardized view of electricity usage over time. The focus on West Bengal allows for a detailed understanding of how consumption patterns have evolved within the state, especially in light of economic development and major disruptions like the COVID-19 pandemic.

The following table includes the summary statistics of this data.

Mean	Median	Standard Deviation	Minimum (2002-03)	Maximum (2019-20)
563.278	578.818	131.506	366.500	756.647

Table 1: Summary Statistics for **Electrical Energy Consumption (in kWh)**

To visually inspect the overall trend and variability in electricity consumption, the following plot illustrates the raw annual per capita consumption data for West Bengal from 2002–03 to 2021–22.

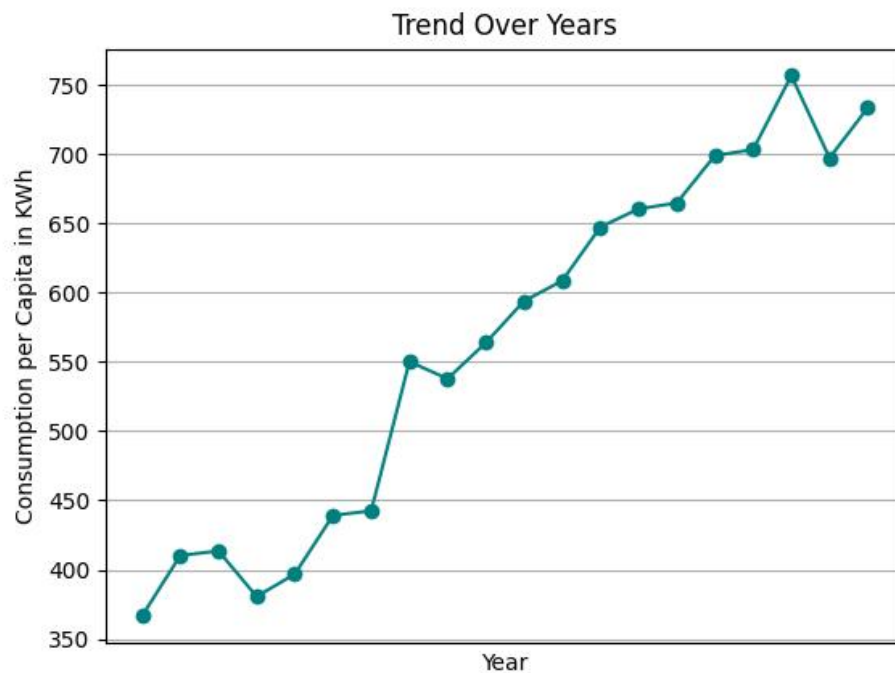


Figure 2.1: Electrical Consumption Plot (2002-2022)

It's worth noting that electricity consumption saw a noticeable decline during the fiscal years 2020–21 and 2021–22, likely due to the disruptions caused by the COVID-19 pandemic.

3 Statistical Analysis

To analyse the temporal patterns in electricity consumption, an additive time series decomposition model was used. This model separates the observed data into three distinct components: trend, seasonal, and random (or error). The additive framework assumes that the overall value at any time is the sum of these three underlying components:

$$Y_t = T_t + S_t + E_t$$

where $\mathbf{Y}_t$ is the observed value at time t , $\mathbf{T}_t$ is the trend component, $\mathbf{S}_t$ is the seasonal component and $\mathbf{E}_t$ is the error component.

A linear trend model is fitted to the annual per capita consumption data spanning the years 2002–03 to 2021–22. The model assumes a relationship of the form:

$$Y_t = \beta_0 + \beta_1 t + \varepsilon_t$$

where Y_t is the observed value at year t , β_0 is the baseline consumption (intercept), β_1 is the annual change (slope), ε_t is the error term and $t \in \{1, 2, \dots, 20\}$ (20 fiscal years). The results are as follows:-

Slope	Intercept	R^2
21.66	335.90	0.9491

Table 2: Trend Model

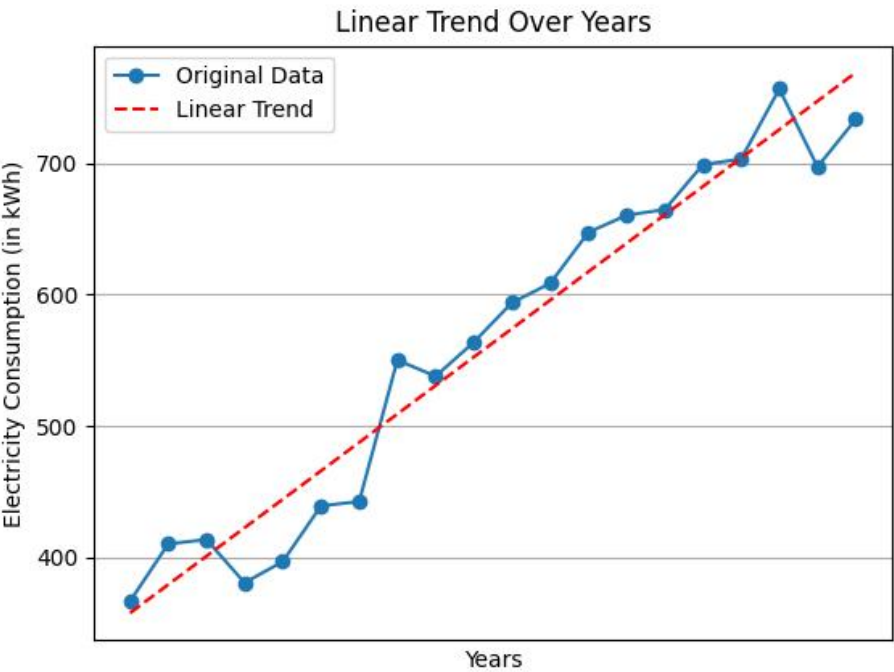


Figure 3.1: Trend Values

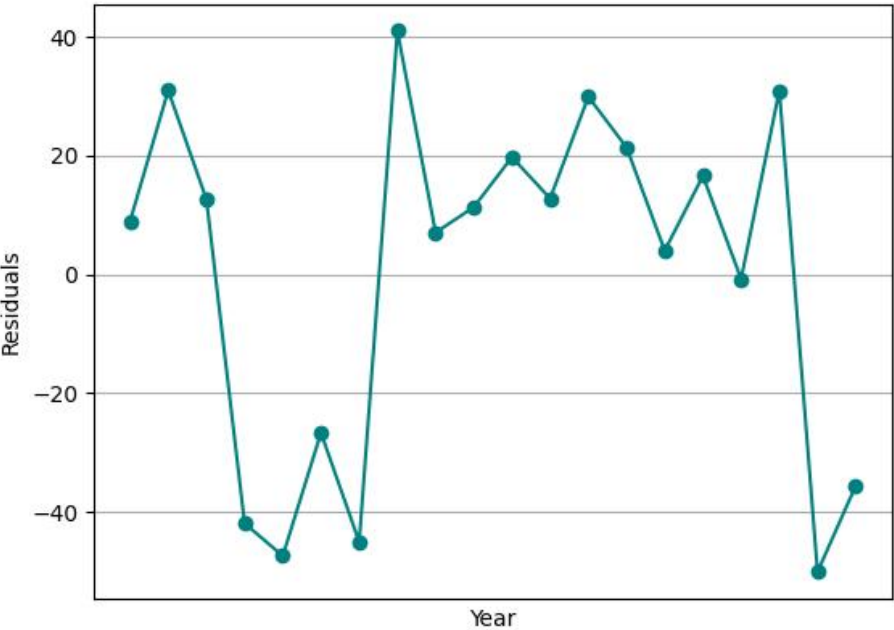


Figure 3.2: Plot of Residuals

When this model is trained on data upto 2019-20 (before COVID), the predicted values are 768.760 kWh (originally 697.266 kWh in 2020-21) and 792.169 kWh (originally 733.418 kWh in 2021-22). Thus, during the pandemic, there is a clear drop in electrical consumption. Again, in the year 2022-23, the model predicts the value to be 815.577 kWh, compared to the actual value which is 819.00 kWh.

In order to account for the effect of COVID, we introduce a dummy variable, C , which takes the value 1 only for the fiscal years 2020-21 and 2021-22 (for $t = 19$ and $t = 20$), otherwise it takes the value 0 for all other values of t . Hence, the new model becomes:

$$Y_t = \beta_0 + \beta_1 t + \beta_2 C + \varepsilon_t$$

The results of fitting this model are as follows:

Slope	Intercept	Coefficient of Dummy Variable	R^2
23.421	323.878	-65.254	0.966

Table 3: Trend Model (with COVID effect)

	2020-21	2021-22	2022-23
(Original)	697.266	733.418	819.000
(Predicted)	703.631	727.052	815.728

Table 4: Comparison between actual and predicted values of Energy Consumption

To evaluate the significance of the model predictors, an ANOVA test was performed on the fitted regression model. This helped assess whether the inclusion of the COVID-19 dummy variable significantly improved the model's explanatory power. The null hypothesis was considered that there is no significant

improvement provided by the COVID model. The ANOVA test returned a p-value of **0.0095**, indicating that the inclusion of the COVID-19 dummy variable significantly improves the model at the **5%** significance level. This suggests that the pandemic period had a statistically meaningful impact on electricity consumption in West Bengal.

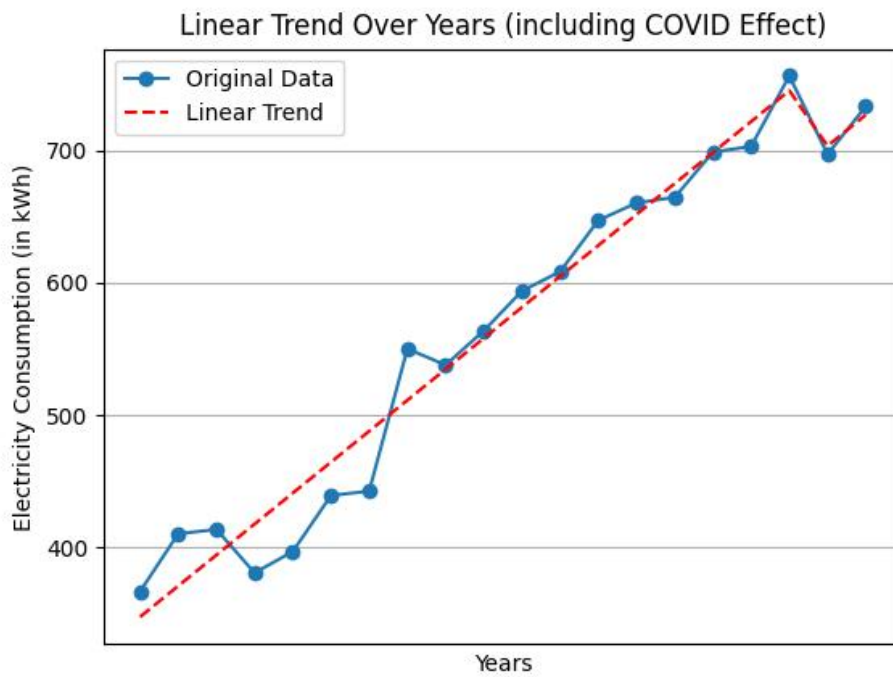


Figure 3.3: Trend Values (with dummy variable)

Not only was there an improvement in the value of R^2 , but the model also predicted the values in the COVID period with great precision. The seasonal component is obtained by subtracting the trend values from the actual observed values. Hence, the time-series is decomposed in the following manner:

Year	Actual (kWh)	Trend (kWh)	Seasonal Component
1	366.500000	347.300000	19.200000
2	410.190000	370.720000	39.470000
3	413.532114	394.140000	19.390000
4	380.610000	417.560000	-36.950000
5	396.790000	440.990000	-44.200000
6	439.170000	464.410000	-25.240000
7	442.450000	487.830000	-45.380000
8	550.159384	511.250000	38.910000
9	537.850056	534.670000	3.180000
10	563.781638	558.090000	5.690000
11	593.855268	581.510000	12.340000
12	608.543126	604.940000	3.610000
13	647.339466	628.360000	18.980000
14	660.469796	651.780000	8.690000
15	664.739001	675.200000	-10.460000
16	698.977064	698.620000	0.360000
17	703.264870	722.040000	-18.780000
18	756.647354	745.460000	11.180000
19	697.265571	703.630000	-6.370000
20	733.418004	727.050000	6.370000

Table 5: Decomposition of Electricity Consumption in West Bengal (2002–2022)

4 Forecasting

Using the fitted linear trend model, forecasts were made for the years beyond the observed data. As shown in the table, the model predicted per capita electricity consumption to be 703.63 kWh in 2020–21, 727.05 kWh in 2021–22, and 815.73 kWh in 2022–23. The actual recorded values for these years were 697.27 kWh, 733.42 kWh, and 819.00 kWh respectively. The differences between the predicted and actual values are relatively small, especially in 2022–23, where the forecast closely matches the real data. These observations indicate that the

model effectively captures the overall consumption trend. The slightly larger deviations during 2020–21 and 2021–22 may be attributed to the temporary disruptions caused by the COVID-19 pandemic. Still, the model’s performance in post-pandemic recovery years suggests it is reliable for short-term forecasting.

5 Inferences

The analysis of electricity consumption in West Bengal over the past two decades, using a simple trend model, reveals some interesting insights. The steady rise in usage reflects growing demand, likely tied to factors like urban expansion, industrial growth, and broader access to electricity in rural areas. This consistent upward trend suggests that the state has been experiencing stable economic progress, with energy needs rising in step.

What stands out is the noticeable drop in consumption during the COVID-19 years. By adding a dummy variable for those years, the model shows just how significant that decline was. This dip suggests that household electricity use alone wasn’t enough to make up for the sharp fall in industrial and commercial demand—highlighting how much of the state’s power consumption comes from those sectors. Still, the quick recovery afterward shows how resilient the state’s economy and energy systems are.

Overall, the strong fit of the model suggests that electricity usage in West Bengal follows a fairly stable pattern, which can be useful for future planning. These findings offer a glimpse into the state’s development journey and how its energy needs are evolving over time.

6 Conclusion

This project explored long-term electricity consumption trends in West Bengal using a linear trend model. The model successfully captured the overall growth in energy demand over a 20-year period, explaining a significant portion of the variation in the data. By incorporating a dummy variable for the COVID-19 years, the model accounted for the sharp decline in usage during that time, highlighting the strong influence of industrial and commercial sectors on overall consumption.

The model's forecasts closely aligned with actual values, particularly in recent years, reinforcing its reliability. While the methodology was intentionally straightforward, the results were meaningful—demonstrating both the state's rising energy needs and its ability to bounce back from temporary disruptions. This simple yet effective modeling approach lays a strong foundation for more detailed future studies and offers valuable direction for energy planning and infrastructure development in the region.

References

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